

STM32 Low-Power Modes and Wake-Up Mechanisms

A technical study of the STM32F103C8T6 ("Blue Pill") power architecture, benchmarked against the ATmega328P (Arduino Uno) sleep-mode model.

Selected device: STM32F103C8T6 ("Blue Pill", Arm® Cortex®-M3)



Selected Device and Reference Sources

STM32F103C8T6

Mainstream Arm® Cortex®-M3 32-bit MCU · STM32F1 series

CORE	Arm Cortex-M3, up to 72 MHz
MEMORY	64 KB Flash, 20 KB SRAM
PACKAGE / BOARD	LQFP48, used on the popular "Blue Pill" board
LOW-POWER MODES	3 operating points: Sleep, Stop, Standby
WHY THIS DEVICE	The most widely used classic STM32 — a simpler, general-purpose power architecture that contrasts with the L4 "ultra-low-power" line

DATASHEET & REFERENCE MANUAL SOURCES

RM0008

Reference Manual — STM32F101/102/103/105/107 advanced Arm-based 32-bit MCUs (DocID13902).
Chapters: PWR, EXTI, RCC, Backup registers.
STMicroelectronics · st.com/resource/en/reference_manual/rm0008...pdf

DS5319

STM32F103x8 / STM32F103xB Datasheet — Mainstream Arm Cortex-M3 32-bit MCU.
STMicroelectronics · st.com/resource/en/datasheet/stm32f103c8.pdf

PM0056

STM32F10xxx Cortex-M3 Programming Manual — WFI / WFE, SCB core registers.
STMicroelectronics / Arm · referenced in RM0008

AN2629

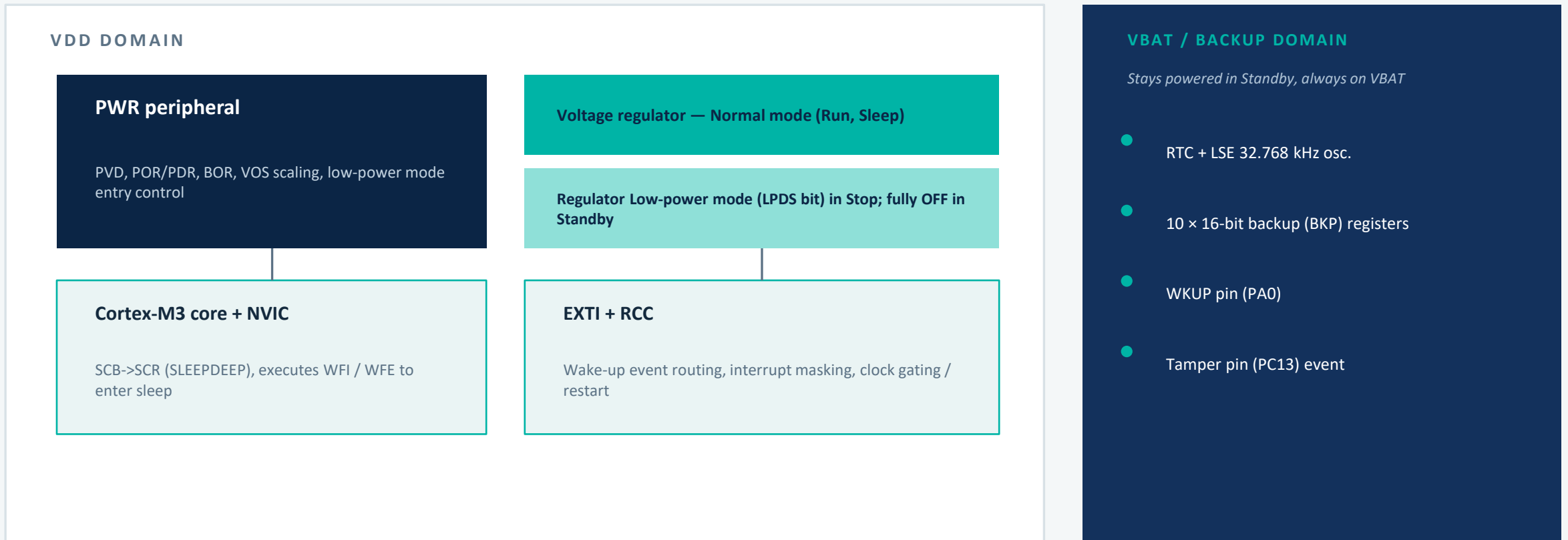
STM32F101xx, STM32F102xx and STM32F103xx Low-power Modes (Application Note).
STMicroelectronics Application Note

ATmega328P

8-bit AVR Microcontroller Datasheet — "Power Management and Sleep Modes" chapter, SMCR register.
Microchip Technology / Atmel

Power-Control Architecture

How the PWR peripheral, voltage regulator, and VBAT/backup domain fit together (RM0008, Ch. 5)



STM32F103 Low-Power Mode Comparison

RM0008 §5.3 — three operating points trade off wake-up latency against retained state

Mode	Regulator	Clocks / core	Retained state	Typical wake-up
Run	Normal mode	All clocks on, CPU running	Full	—
Sleep	Normal mode	CPU clock off, peripherals run	Full	Immediate (WFI/WFE)
Stop	Normal or Low-power (LPDS bit)	HSI/HSE/PLL stopped, 1.8 V domain clocks halted	SRAM + registers	~μs (longer if LPDS = 1)
Standby	Off (1.8 V domain unpowered)	Everything off except VBAT domain	Backup (BKP) registers only	Reset-like, full reboot

Unlike the STM32L4 "ultra-low-power" line, the F1 mainstream series has no Low-power Run, no graduated Stop 0/1/2 tiers, and no Shutdown mode — Standby is the single deepest state.

Important Registers

The register set that configures, enters, and reports on low-power operation

1

PWR_CR

Selects Stop vs. Standby on deep-sleep entry (PDDS bit), regulator mode in Stop (LPDS bit), and PVD threshold (PLS bits).

2

PWR_CSR

Status/config: enables the WKUP pin (EWUP), and reports PVD output, Standby flag (SBF), and wake-up flag (WUF).

3

RCC_BDCR

Backup domain control: LSE oscillator enable, RTC clock source select, RTC enable — independent of the core's sleep state.

4

BKP_DR1...DR10

Ten 16-bit backup data registers, powered from VBAT and retained through Standby (SRAM itself is lost).

5

SCB->SCR

Cortex-M3 core register: SLEEPDEEP selects Sleep vs. deep-sleep (Stop/Standby); SLEEPONEXIT auto-re-enters sleep after an ISR.

6

EXTI_IMR / RTSR / FTSR / PR

Interrupt mask, rising/falling edge selection, and pending flags for the 16 GPIO wake-up lines plus internal lines (RTC, USB).

Important Register Bits

Key bit fields used when writing low-power / wake-up firmware

Register	Bit field	Function
PWR_CR	PDDS	0 = deep-sleep enters Stop mode; 1 = deep-sleep enters Standby mode
PWR_CR	LPDS	Selects the regulator state during Stop: 0 = normal, 1 = low-power
SCB_SCR	SLEEPDEEP	0 = Sleep mode on WFI/WFE; 1 = enter the mode selected by PWR_CR.PDDS
SCB_SCR	SLEEPONEXIT	Re-enters sleep automatically when the last active ISR returns
PWR_CR	CWUF / CSBF	Clears the WUF wake-up flag / the SBF Standby flag before re-entering sleep
PWR_CSR	EWUP / WUF / SBF	Enables the WKUP pin (PA0); reports the wake-up source / prior Standby exit
EXTI_IMR	MRx	Unmasks interrupt line x so it can wake the CPU (used with WFI)
EXTI_RTSR / FTSR	TRx	Selects rising- and/or falling-edge sensitivity for each EXTI line

WFI / WFE and the Wake-Up Mechanism

PM0056 — the two Cortex-M3 instructions used to enter sleep, and how a wake-up event reaches the core

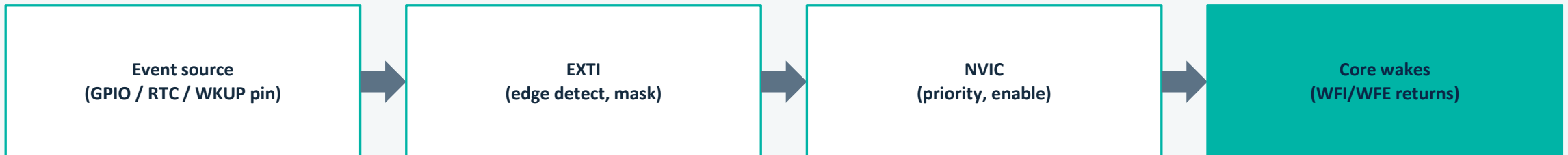
WFI — Wait For Interrupt

- Sleeps until any interrupt enabled in the NVIC occurs
- The matching ISR always runs on wake-up
- Simplest, most common way to enter Sleep/Stop/Standby

WFE — Wait For Event

- Wakes on an EXTI event or the internal event register (SEV)
- Does not require the interrupt to be unmasked in the NVIC
- Useful for low-overhead polling loops with no ISR needed

WAKE-UP EVENT PATH



Practical Applications

Where STM32 low-power modes matter in real products

1

Hobbyist & prototyping boards

Sleep mode between polling loops on "Blue Pill" projects; Stop mode extends runtime in simple battery-powered builds.

2

USB & CAN peripherals

Sleep keeps USB/CAN clocks alive while the CPU idles between packet or frame processing — no data is missed.

3

Motor & robotics controllers

Stop mode between control-loop iterations on battery-powered robots; EXTI wakes the CPU on the next encoder or sensor edge.

4

Cost-sensitive battery products

Standby with a WKUP-pin (PA0) button press for simple always-off-until-pressed devices such as alarms or remote sensors.

ATmega328P Sleep-Mode Model

The 8-bit AVR reference point — as used on the Arduino Uno

SMCR — Sleep Mode Control Register



SM[2:0] select the mode; SE (Sleep Enable) must be set to 1 before the SLEEP instruction is executed.

SIX SLEEP MODES

- 1 Idle
- 2 ADC Noise Reduction
- 3 **Power-down (deepest)**
- 4 Power-save
- 5 Standby
- 6 Extended Standby

WAKE-UP SOURCES DEPEND ON MODE DEPTH

Idle: any enabled interrupt (Timer, USART, SPI, ADC, INT0/1, PCINT).

Power-down: external reset, Watchdog Timeout, INT0/INT1 (level), pin-change interrupt, TWI address match.

Power-save / Standby: same as Power-down, plus Timer/Counter2 asynchronous (external 32.768 kHz crystal) events.

Two Different Power-Management Philosophies

ATmega328P

- One register (SMCR) selects among 6 fixed sleep modes
- No dedicated real-time clock , Timer2 async mode is the closest substitute
- RAM is retained in every sleep mode as long as VCC is present
- Simple, predictable, low cost , ideal for small, single-task designs

STM32F103

- A single PWR peripheral offers 3 modes (Sleep, Stop, Standby), set with just two bits , LPDS and PDDS
- Dedicated RTC in the always-on VBAT domain, independent of core sleep state, backed by only 10 backup registers
- RAM retention is binary: fully retained in Sleep/Stop, entirely lost in Standby , no partial-SRAM tier like the L4's SRAM2
- Simpler than the ultra-low-power L4 line, but still far more flexible than the ATmega328P , a solid general-purpose mainstream MCU

Final Comparison: ATmega328P vs. STM32

Feature	ATmega328P	STM32 (F103C8T6)
Sleep modes	6: Idle, ADC NR, Power-down, Power-save, Standby, Ext. Standby	3: Sleep, Stop, Standby (no LP Run / graduated Stop tiers)
Deepest low-power mode	Power-down (~ μ A range)	Standby (μ A range; only VBAT domain retained, SRAM lost)
Sleep-control registers	SMCR (SM2:0, SE)	PWR_CR (LPDS, PDDS), SCB_SCR (SLEEPDEEP)
Wake-up sources	INT0/1, PCINT, WDT, TWI addr. match, Timer2, ADC	EXTI lines, WKUP pin (PA0), RTC alarm, NRST, IWDG reset
External interrupt wake-up	INT0 / INT1 + pin-change (PCINT) in all modes	16 EXTI lines via NVIC in Sleep & Stop; Standby wakes via WKUP only
Timer operation during sleep	Timer2 only, async 32.768 kHz (Power-save)	All timers stop in Stop mode (no dedicated low-power timer)
RTC operation	None dedicated (Timer2 as substitute)	Dedicated RTC in backup domain, active through Stop and Standby
RAM retention	Retained in all sleep modes (VCC present)	Full in Sleep/Stop; entirely lost in Standby (backup regs. only)
Typical applications	Simple battery devices, hobbyist sensors, always-on control	General embedded control, motor/robotics, USB/CAN, budget prototyping

Key Takeaways

1

The STM32F103's PWR peripheral offers a compact three-mode model , Sleep, Stop, Standby , controlled by just the LPDS and PDDS bits, still far more configurable than the ATmega328P's six fixed modes.

2

WFI and WFE, combined with SCB->SCR.SLEEPDEEP and PWR_CR.PDDS, are the two entry points into every STM32 low-power state , the wake-up path runs through EXTI and the NVIC in Sleep and Stop; Standby instead restarts like a reset.

3

The ATmega328P's simplicity is a genuine strength for small, single-task, cost-sensitive designs, while the STM32F103's dedicated always-on RTC and 72 MHz Cortex-M3 core suit general-purpose products that need more compute headroom with modest power savings.

4

Correct register sequencing , clearing wake-up flags, disabling unused peripheral clocks, and choosing the shallowest mode that meets the latency budget , is what actually delivers the datasheet current figures.

Sources Cited

- [1] STMicroelectronics. RM0008: STM32F101/102/103/105/107 Advanced Arm®-Based 32-Bit MCUs — Reference Manual, DocID13902. https://www.st.com/resource/en/reference_manual/rm0008-stm32f101xx-stm32f102xx-stm32f103xx-stm32f105xx-and-stm32f107xx-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
- [2] STMicroelectronics. STM32F103x8 / STM32F103xB Datasheet — Mainstream Arm® Cortex®-M3 32-Bit MCU, DS5319. <https://www.st.com/resource/en/datasheet/stm32f103c8.pdf>
- [3] STMicroelectronics / Arm. PM0056: STM32F10xxx Cortex®-M3 Programming Manual (WFI/WFE, SCB core registers).
- [4] STMicroelectronics. AN2629 — STM32F101xx, STM32F102xx and STM32F103xx Low-Power Modes, Application Note. https://www.st.com/resource/en/application_note/an2629-stm32f101xx-stm32f102xx-and-stm32f103xx-lowpower-modes-stmicroelectronics.pdf
- [5] Microchip Technology / Atmel. ATmega328P — 8-Bit AVR Microcontroller Datasheet, "Power Management and Sleep Modes" chapter (SMCR register).

All device data reflects the STM32F103x8 / ATmega328P documentation current as of the sources listed above.